



Application Layer Protocols

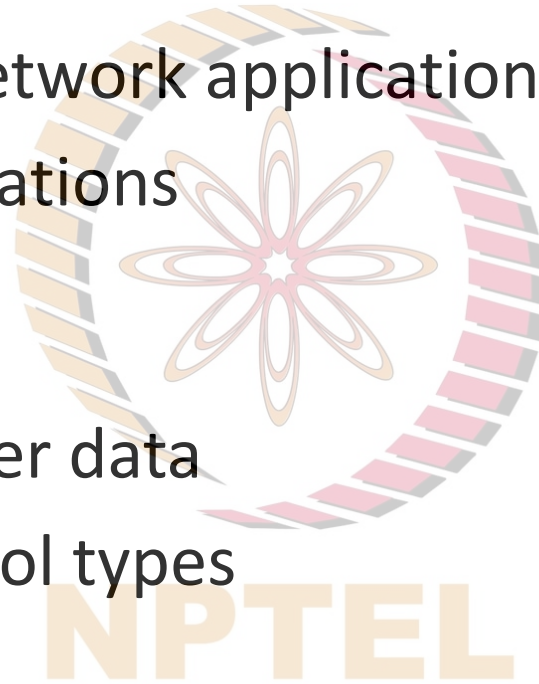
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Overview

- ☐ Understand what is a network application
- ☐ Types of network applications
 - ☐ Client- Server model
 - ☐ Peer-to-Peer model
- ☐ How applications transfer data
- ☐ Application layer protocol types
 - ☐ Text based
 - ☐ Binary

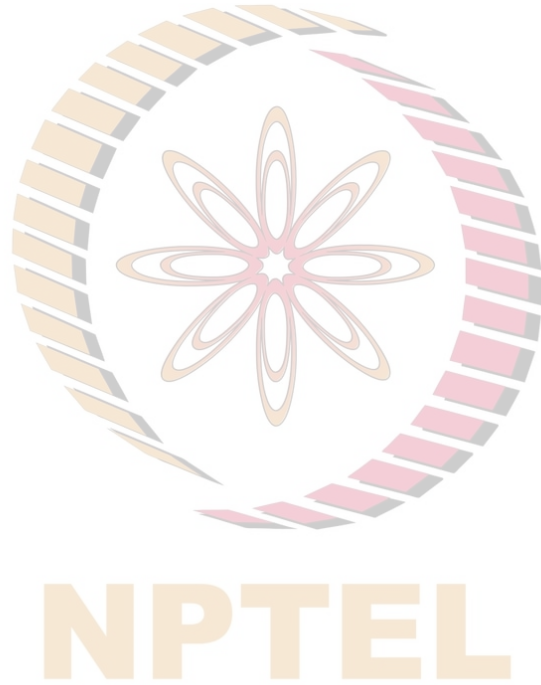


Types of Programs

- ❑ Software programs come in 3 main variants
 - ❑ **Application program:** These are the programs/applications that the end user uses/interacts with
 - ❑ **System program:** Usually manages the hardware and resources on the system. Facilitates execution of application programs
 - ❑ **Utility program:** Programs used for configuration and maintenance of system

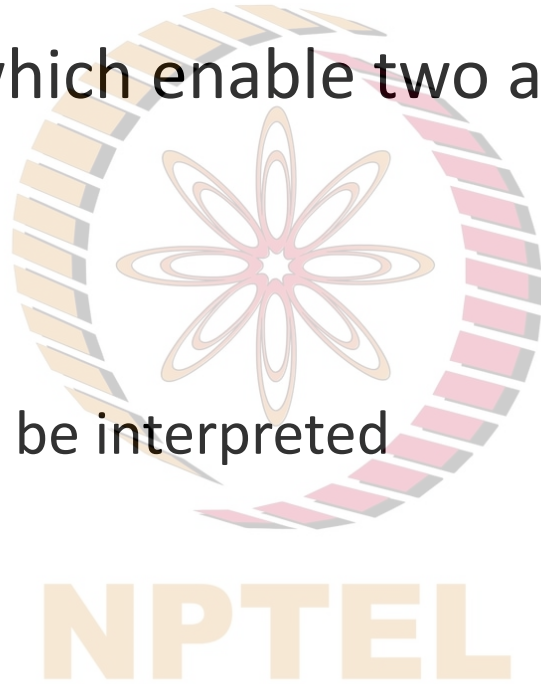
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What is a Network Application ?



What are Application Layer Protocols ?

- ☐ It is a type of protocol which enable two applications exchange data over network.
- ☐ This protocol tells
 - ☐ how data to be sent
 - ☐ How data received has to be interpreted



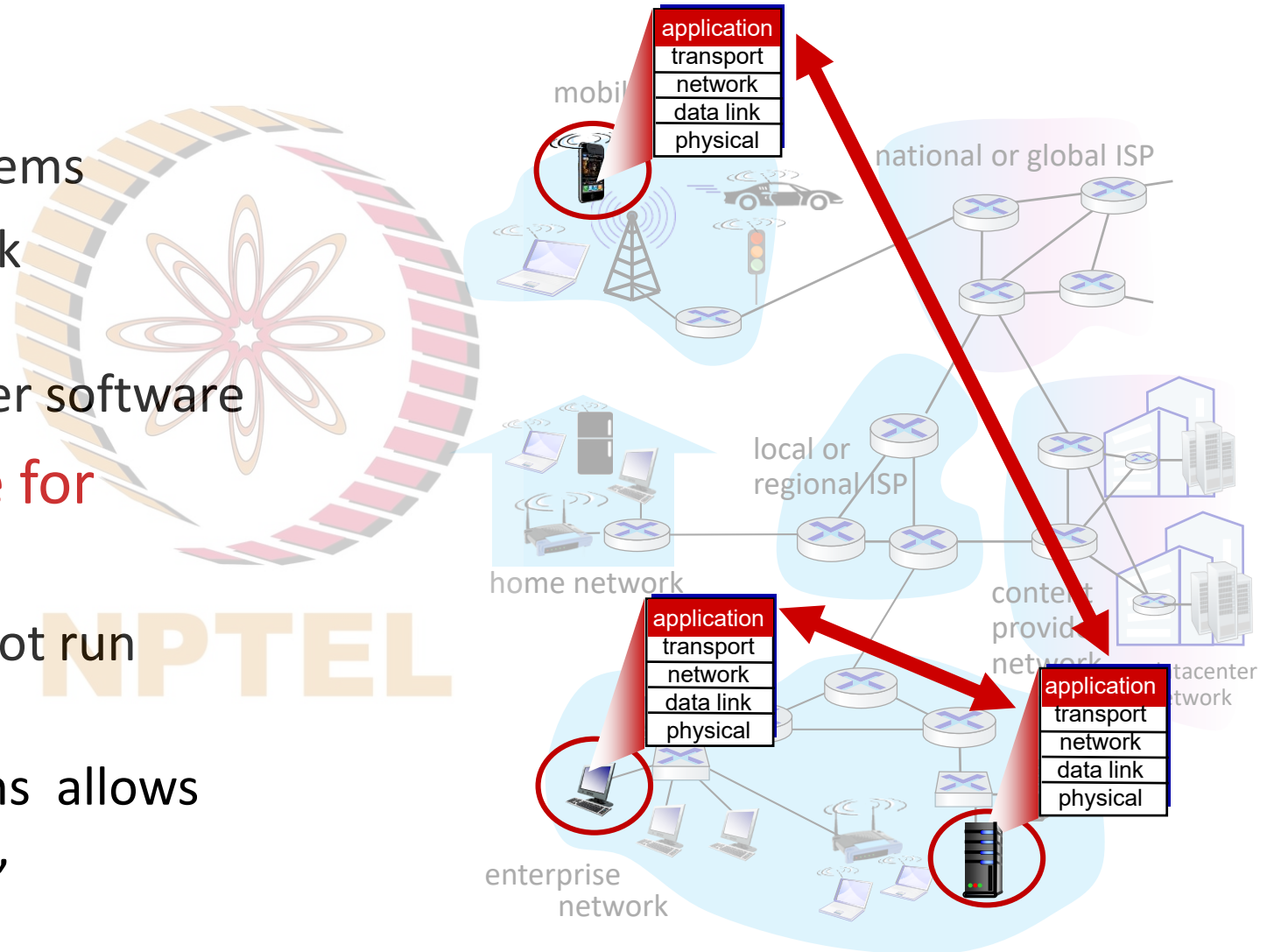
Creating a Network Application

Write programs that:

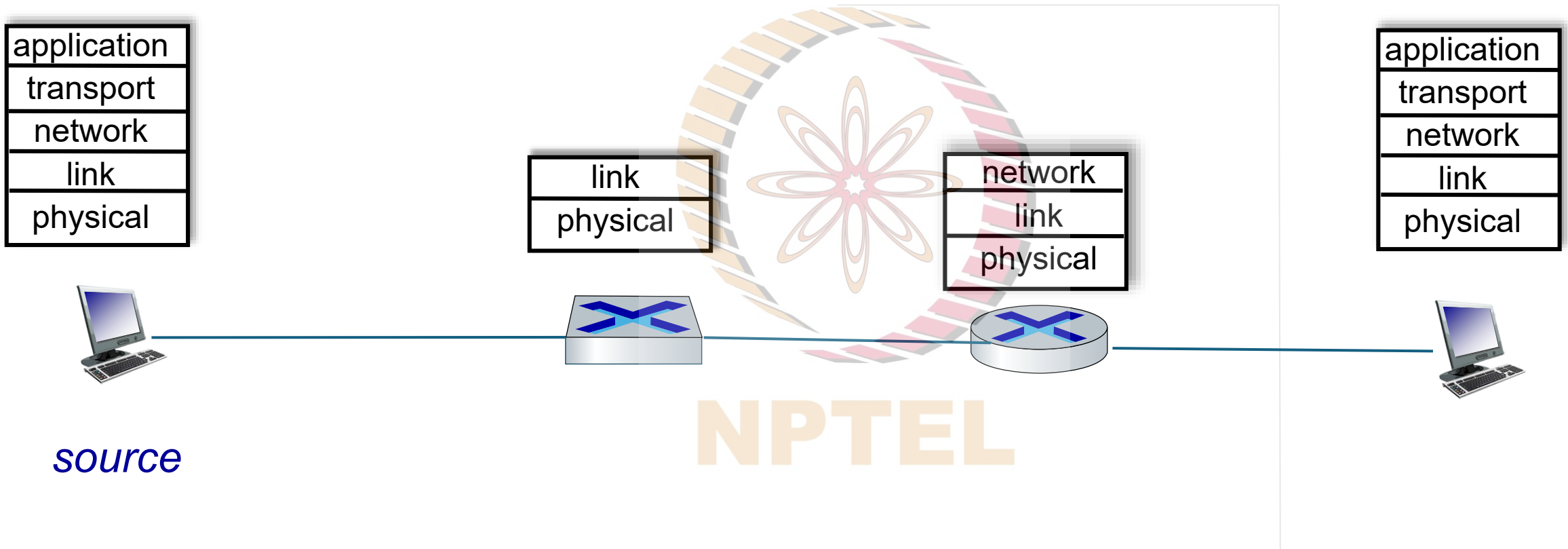
- ❑ Run on (different) end systems
- ❑ Communicate over network
- ❑ e.g., web server software communicates with browser software

No need to write software for network-core devices

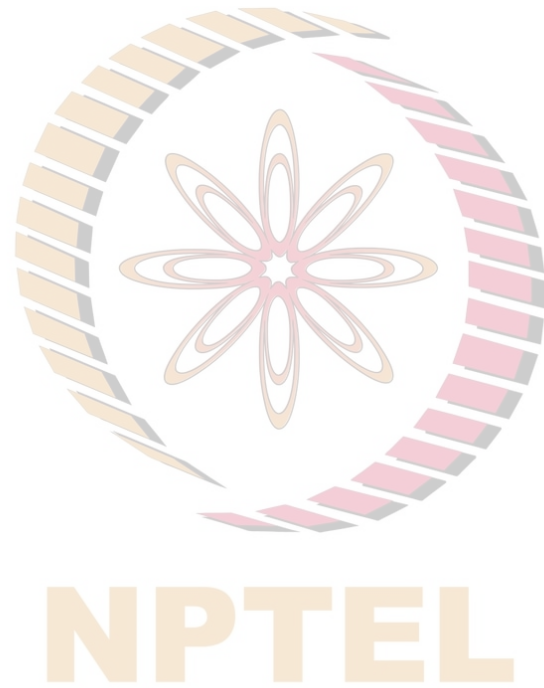
- ❑ Network-core devices do not run user applications
- ❑ Applications on end systems allows for rapid app development, propagation



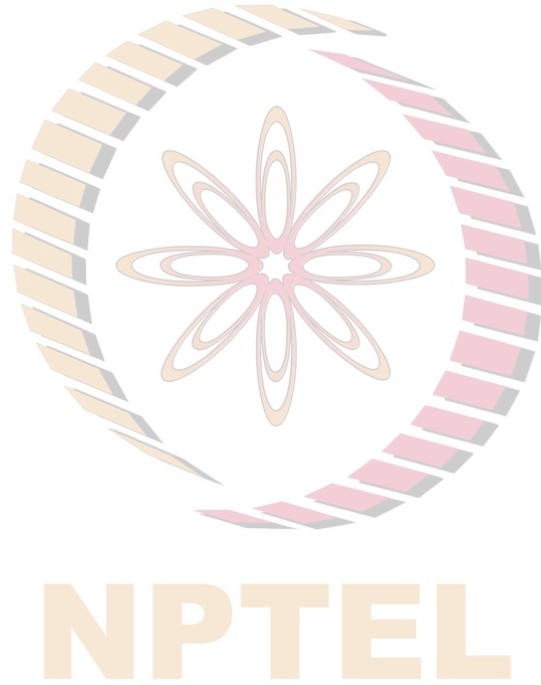
Application Development



Client Server Architecture



Peer-to-Peer Architecture



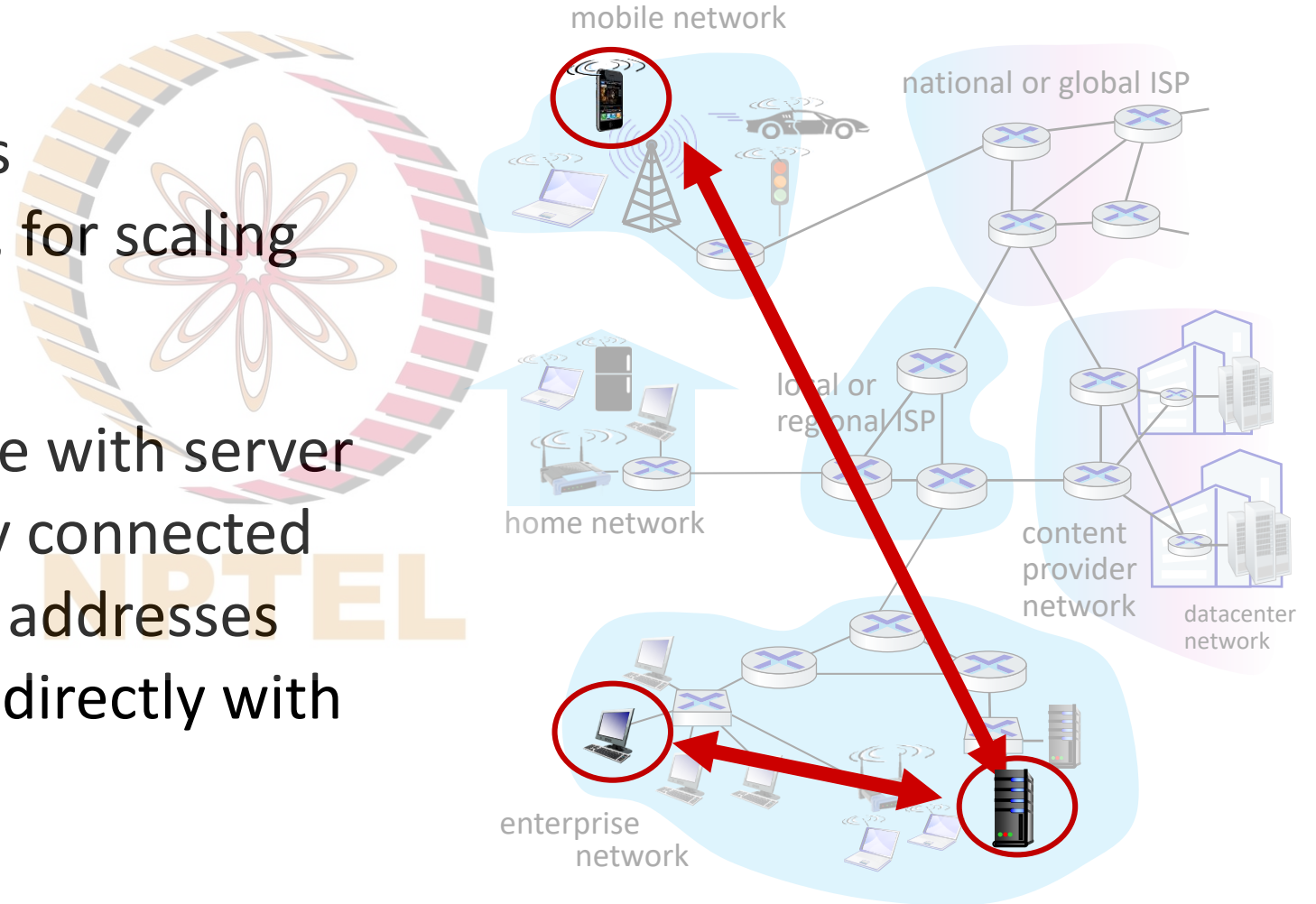
Client-Server Paradigm

Server:

- ❑ Always-on host
- ❑ Permanent IP address
- ❑ Often in data centers, for scaling

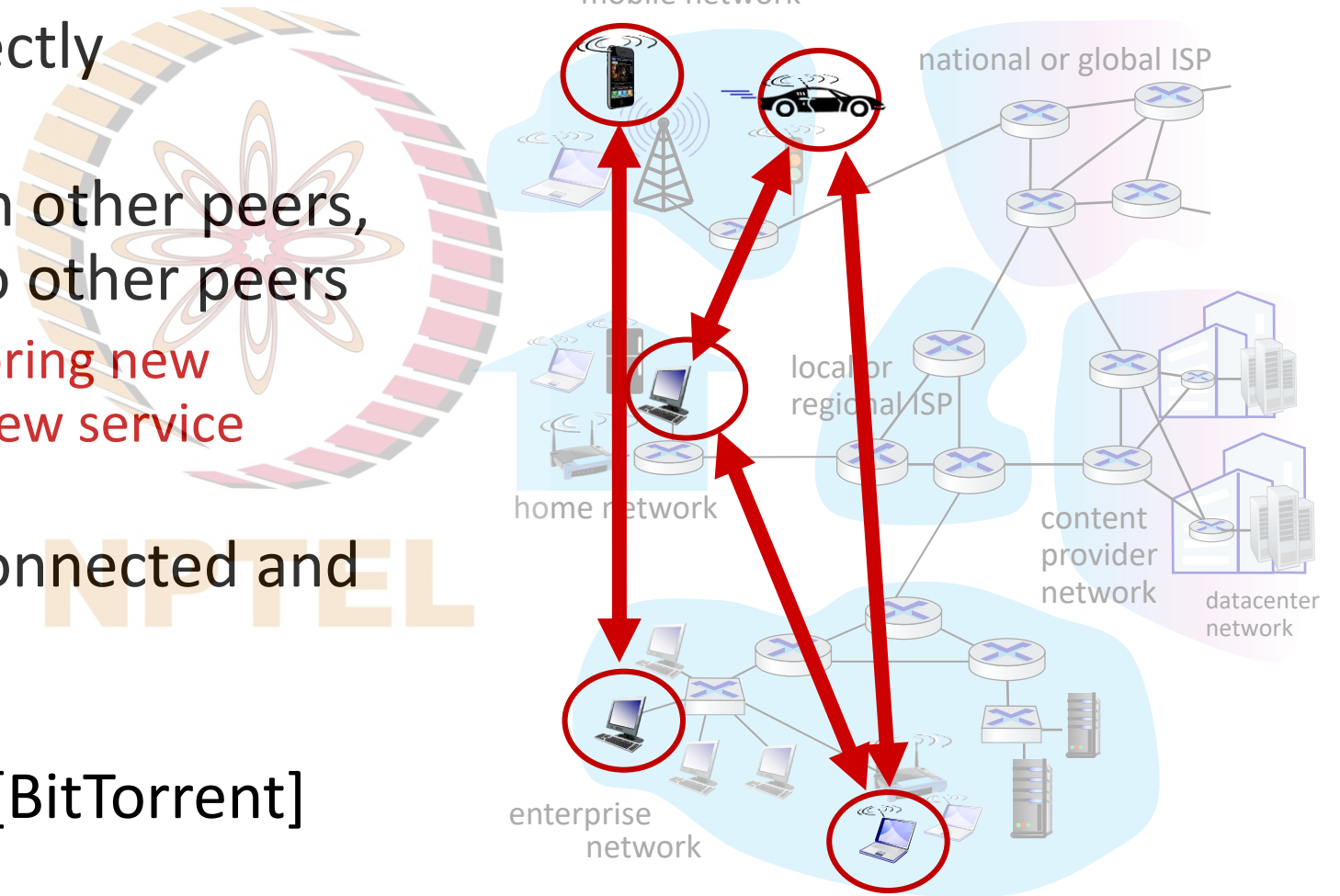
Clients:

- ❑ Contact, communicate with server
- ❑ May be intermittently connected
- ❑ May have dynamic IP addresses
- ❑ Do *not* communicate directly with each other
- ❑ Ex: HTTP, IMAP, FTP



Peer-peer Architecture

- ❑ No always-on server
- ❑ Arbitrary end systems directly communicate
- ❑ Peers request service from other peers, provide service in return to other peers
 - *self scalability* – new peers bring new service capacity, as well as new service demands
- ❑ Peers are intermittently connected and change IP addresses
 - ❑ complex management
- ❑ example: P2P file sharing [BitTorrent]



Client Server v/s P2P Traffic in 2026

☐ **Client-Server:** The Massive Majority (~95%+)

- ☐ Video & Cloud Streaming
- ☐ AI inference traffic
- ☐ Bot Traffic - 51% of web traffic is now generated by bots (crawlers, scrapers, and automated tools)

☐ **Peer-to-Peer (P2P):** The Strategic Niche (~3–5%)

- ☐ Hybrid P2P CDNs
- ☐ P2P Payments & Finance

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